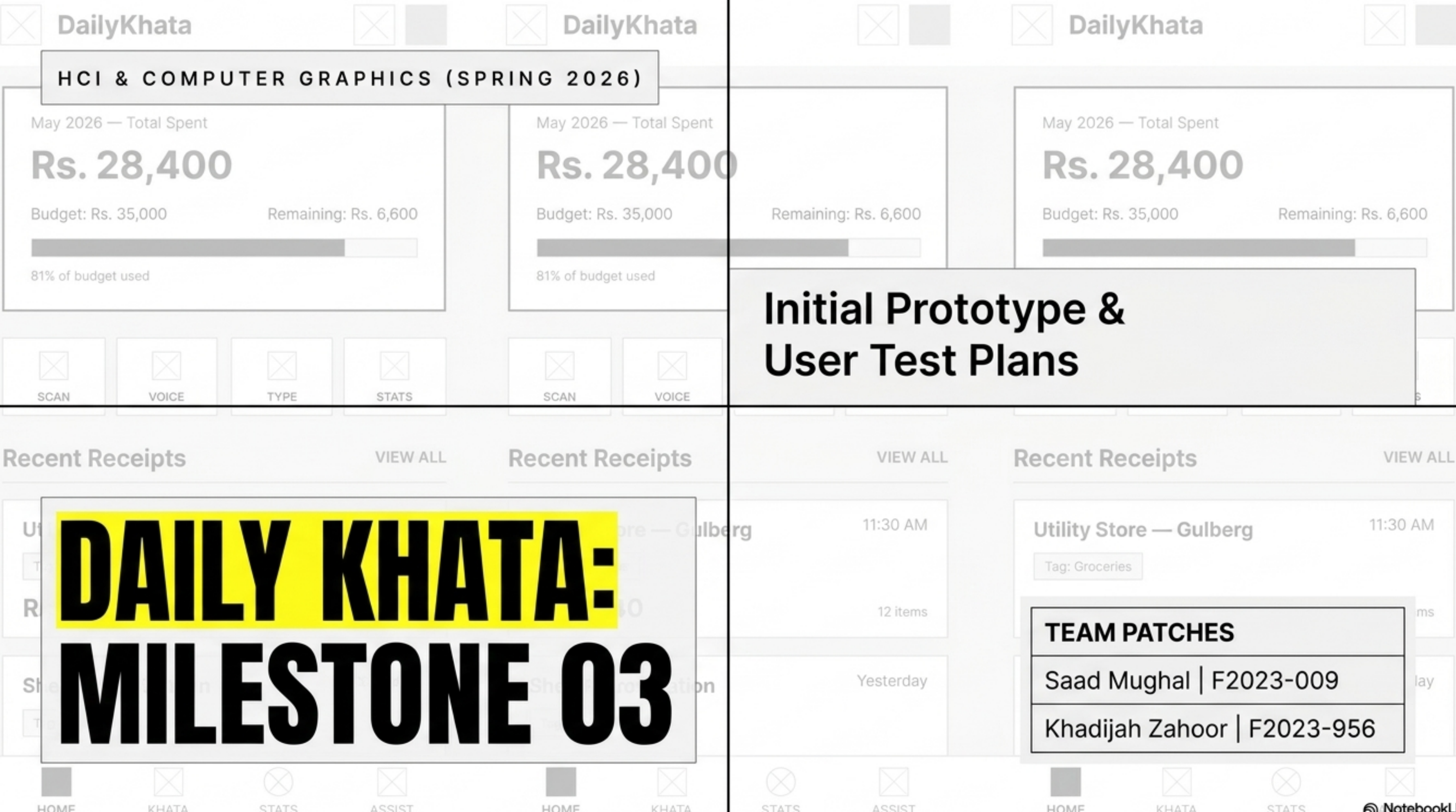




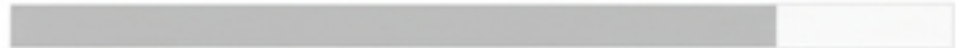
HCI & COMPUTER GRAPHICS (SPRING 2026)

May 2026 — Total Spent

Rs. 28,400

Budget: Rs. 35,000

Remaining: Rs. 6,600



81% of budget used

May 2026 — Total Spent

Rs. 28,400

Budget: Rs. 35,000

Remaining: Rs. 6,600



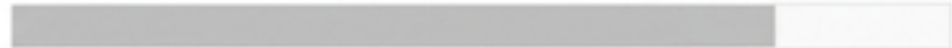
81% of budget used

May 2026 — Total Spent

Rs. 28,400

Budget: Rs. 35,000

Remaining: Rs. 6,600



Initial Prototype &
User Test Plans

Recent Receipts

VIEW ALL

Recent Receipts

VIEW ALL

Recent Receipts

VIEW ALL

DAILY KHATA:
MILESTONE 03

Utility Store — Gulberg

11:30 AM

Tag: Groceries

11:30 AM

12 items

Yesterday

TEAM PATCHES

Saad Mughal | F2023-009

Khadijah Zahoor | F2023-956

HOME

KHATA

STATS

ASSIST

HOME

KHATA

STATS

ASSIST

HOME

KHATA

STATS

ASSIST

SECTION 01

PROTOTYPE DESIGN & EXECUTION

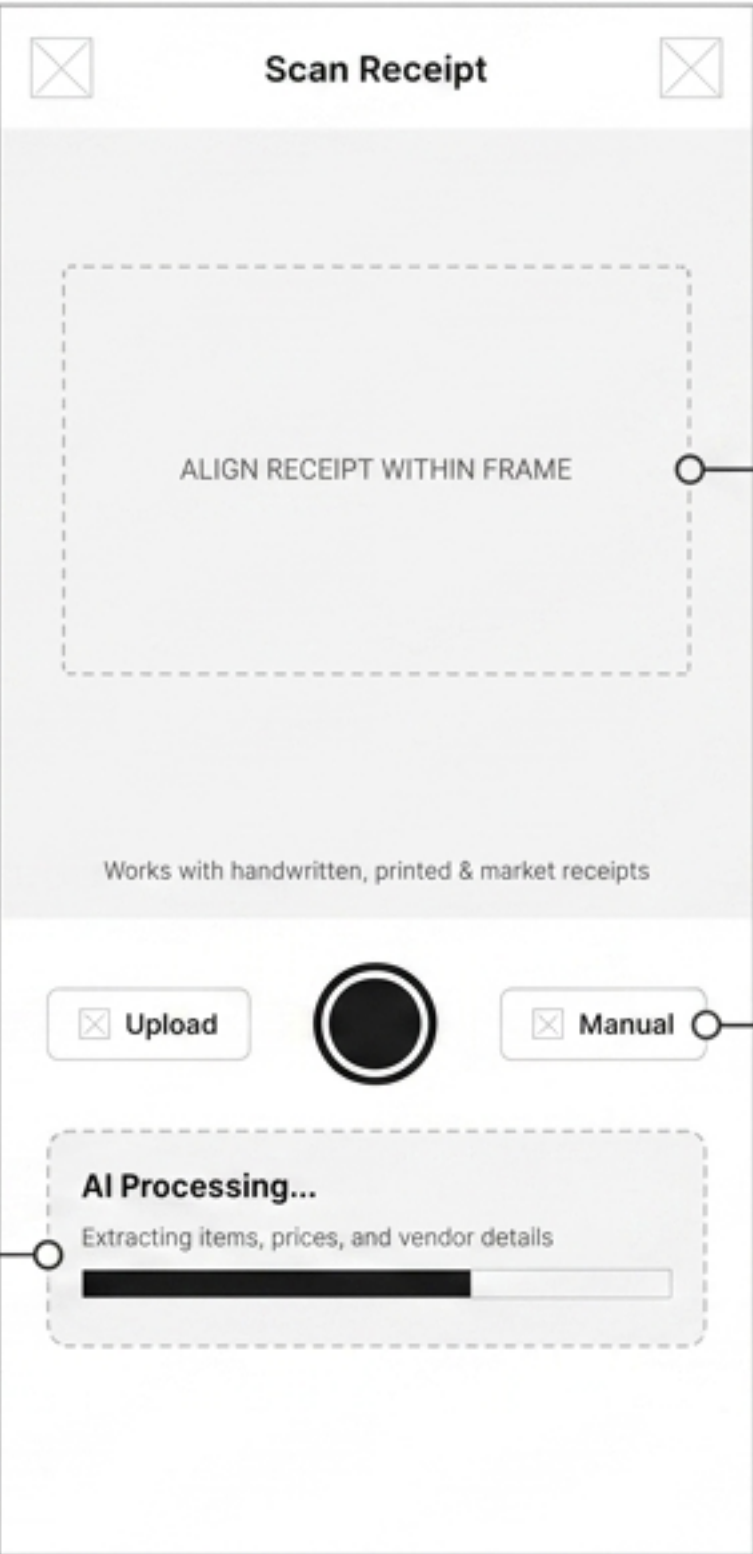
Translating user needs into a low-fidelity, fully interactive, offline-first architecture.

PROTOTYPE SCOPE & FIDELITY

PROTOTYPE GOALS	INCLUSION / EXCLUSION	DATA CONTEXT
● Evaluate AI receipt verification pipeline. ● Assess comprehension of Analytics/Price History. ● Test multimodal entry (Scan/Voice/Manual).	IN SCOPE: ● 6 primary task flows ● Interactive error paths ● Low-fidelity grayscale UI (focus on layout/logic over aesthetics) OUT OF SCOPE: ● Live OCR processing ● Cloud syncing ● Account creation	Hardcoded Pakistani-context dummy data to simulate realistic usage. ● Rs. 28,400 monthly spend ● Utility Store, Atta, LESCO

PROTOTYPE LINK: Active Figma prototype bridging all 13 core screens into a cohesive experience.

UI ANATOMY: MULTIMODAL FRICTIONLESS INPUT



Viewfinder designed specifically for handwritten/faded market receipts.

Fallback 'Manual' entry integrated at root level to prevent dead ends.

Explicit 'AI Processing' state manages user expectations regarding latency.



'Listening' badge ensures system status visibility.

Live transcript builds system trust.

AI-detected item cards populate in real-time, allowing instant verification before saving.

UI ANATOMY: MANDATORY INLINE VERIFICATION

DESIGN RATIONALE

Users fundamentally distrust AI with financial data.

Auto-commit is dangerous; it is replaced with a mandatory review step to ensure data integrity.



Verify Receipt

Save

Utility Store — Gulberg, Lahore

Receipt Date: 16 May 2026 - 11:30 AM

1000 · Food & Groceries

ITEM	QTY	UNIT	TOTAL
Atta (10kg)	1	212	Rs. 212
Doodh (1L)	4	180	Rs. 720
Chawal (5kg)	1	340	Rs. 340
Cooking Oil	2	480	Rs. 960
Total			Rs. 2,232

3 fields flagged — tap to correct before saving

Price Changes Since Last Visit

Atta: Rs. 180 → Rs. 212 +18%

Doodh: Rs. 175 → Rs. 180 +3%

CONFIRM & SAVE TO KHATA

EDIT ITEMS

UNCERTAINTY FLAGS

Fields with low AI confidence are explicitly highlighted with a dashed yellow border.

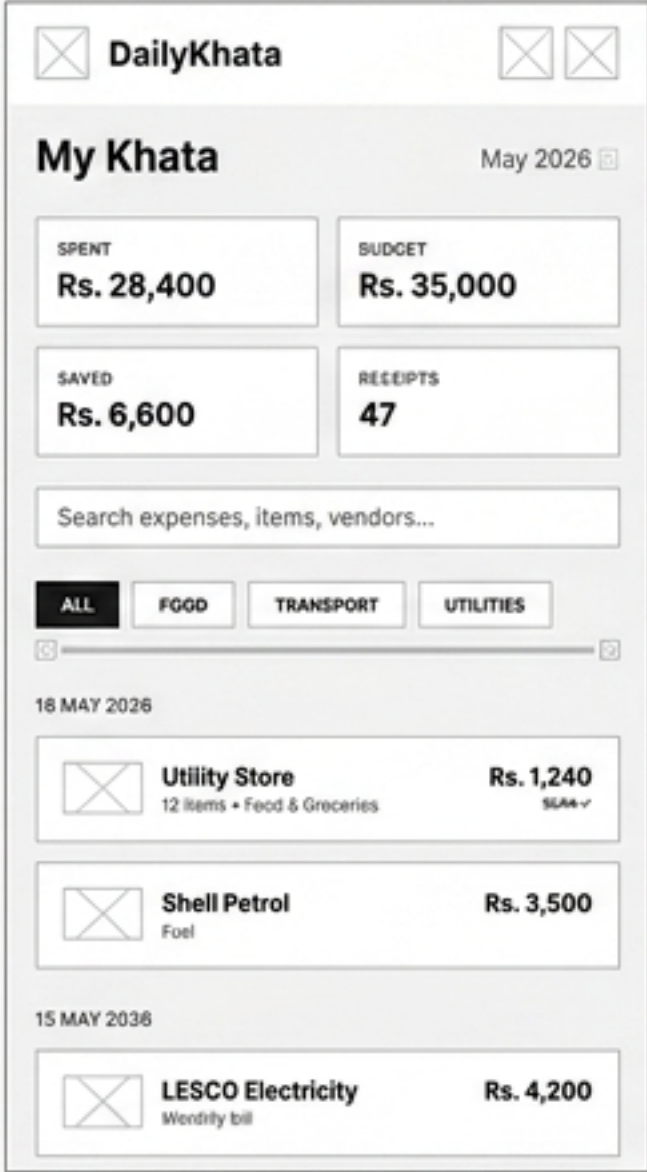
TOUCH-TO-EDIT

Tapping any flagged field allows instant correction via a numpad, providing rapid error recovery.

CONTEXTUAL ANCHORS

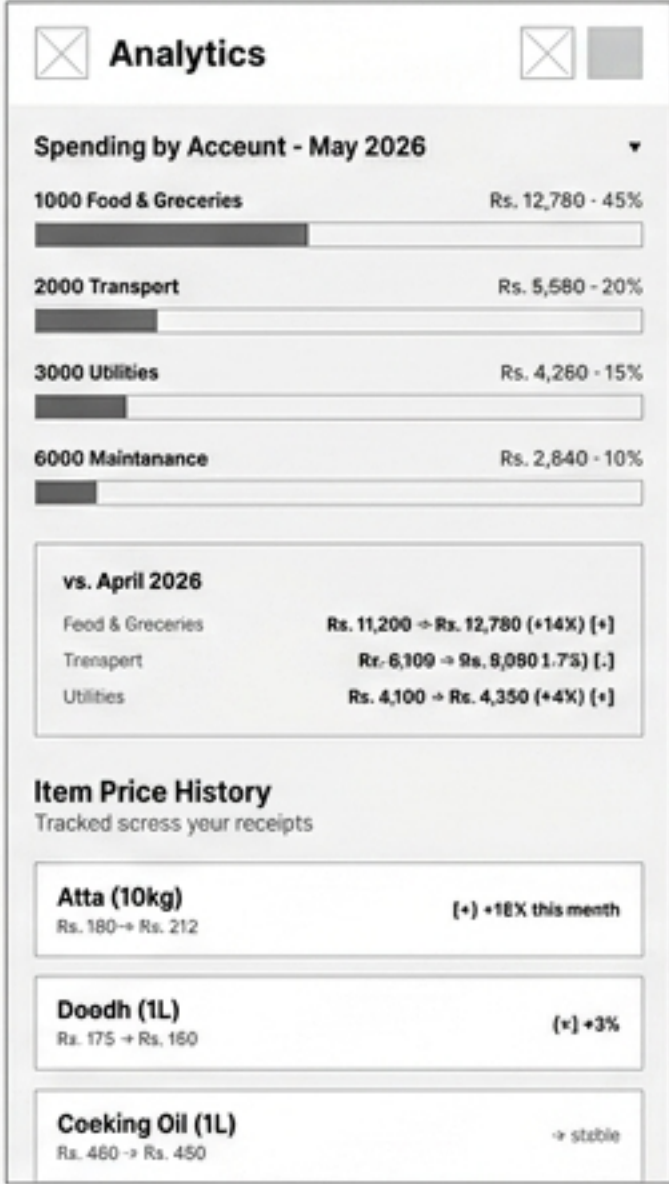
Month-over-month price changes (e.g., Atta +18%) are displayed directly on the verification screen to contextualize spending immediately.

UI ANATOMY: LEDGER, ANALYTICS, AND CONVERSATIONAL QUERY



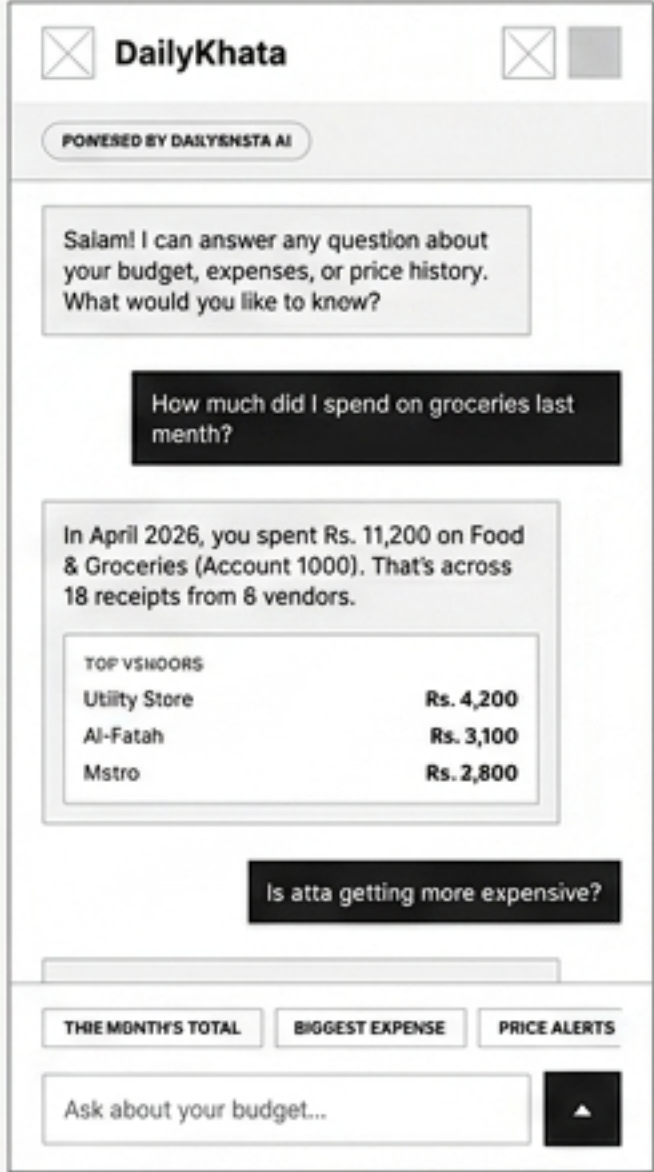
DIGITAL KHATA [WF-04]

A full digital ledger featuring horizontal filter pills (Food, Transport, Utilities) and explicit scan verification badges per entry for traceability.



ANALYTICS [WF-05]

Avoids complex charts in favor of plain-language month-over-month changes and a dedicated item price tracker (e.g., Atta price history).



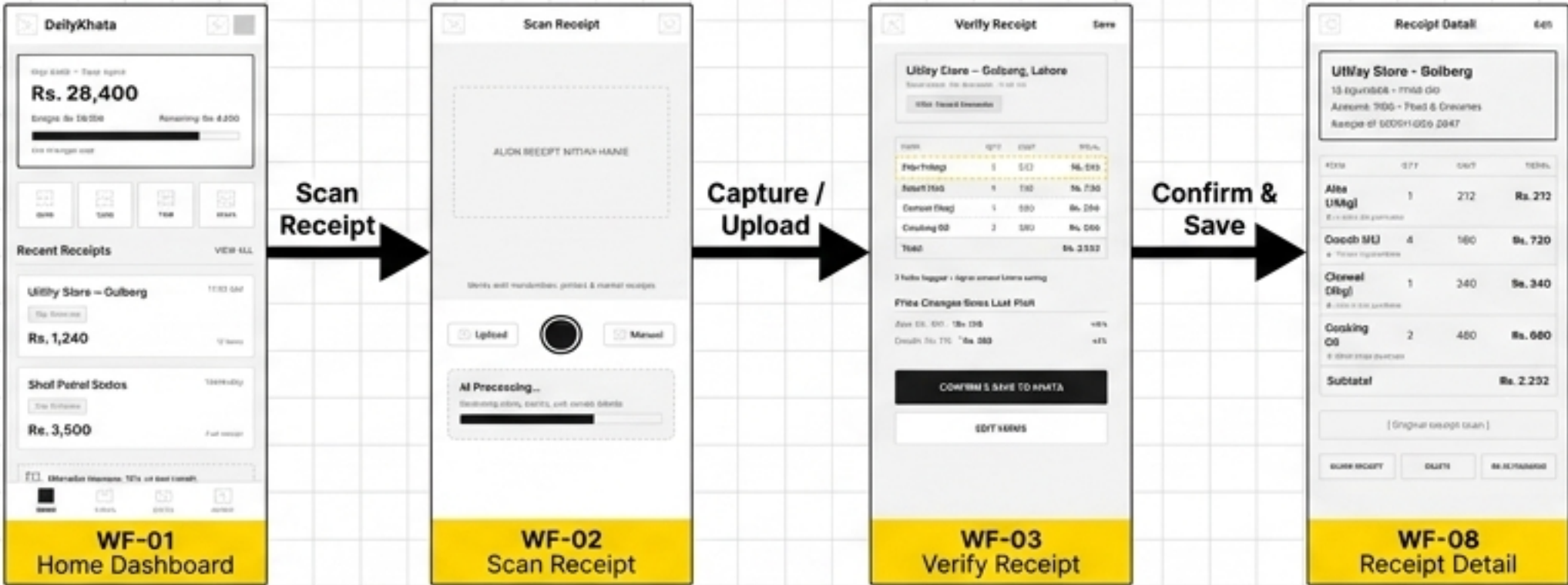
AI ASSISTANT [WF-06]

A conversational interface for complex queries. Includes predefined quick-question pills (e.g., 'Biggest expense') to support users with lower digital literacy.

CORE NAVIGATION: RECEIPT SCANNING & MANUAL ENTRY

FLOW A: SCAN & VERIFY (Task T1)

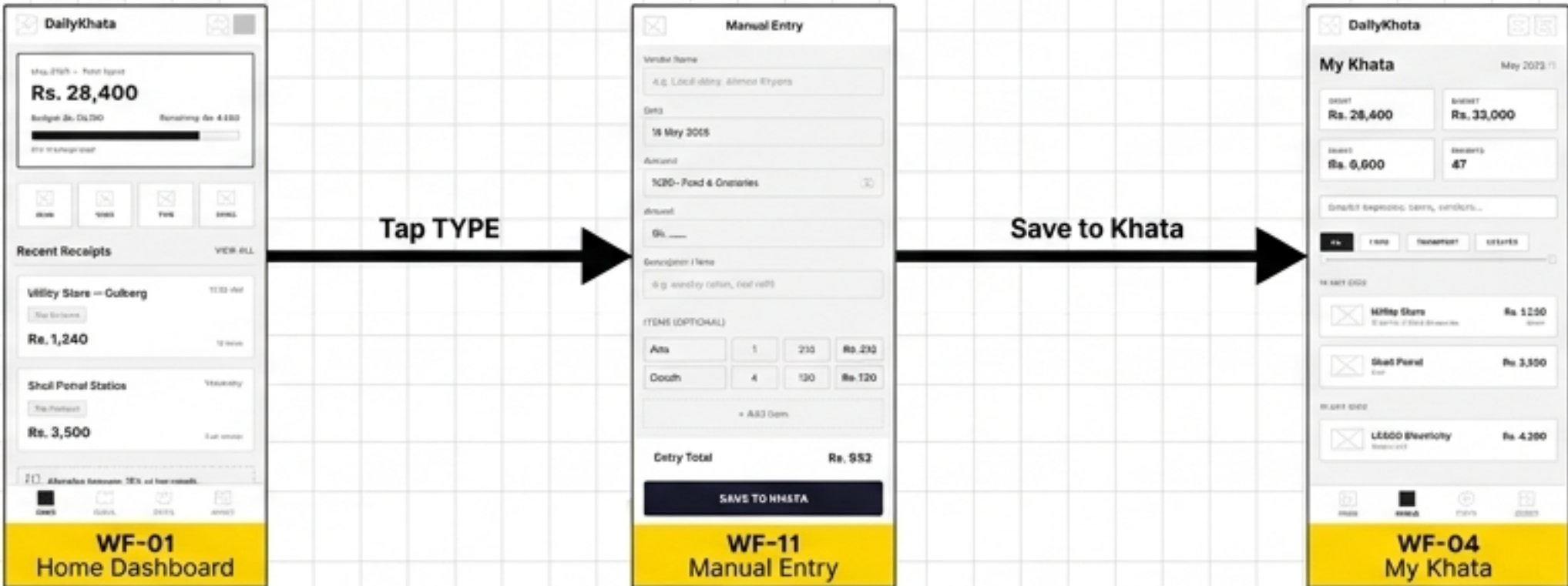
Home Dashboard -> Capture/Upload
-> Verify Flags -> Confirm & Save to
Detail View.



FLOW C: MANUAL ENTRY (Task T3)

Designed to accommodate informal,
receipt-less cash transactions
(e.g., paying a plumber).

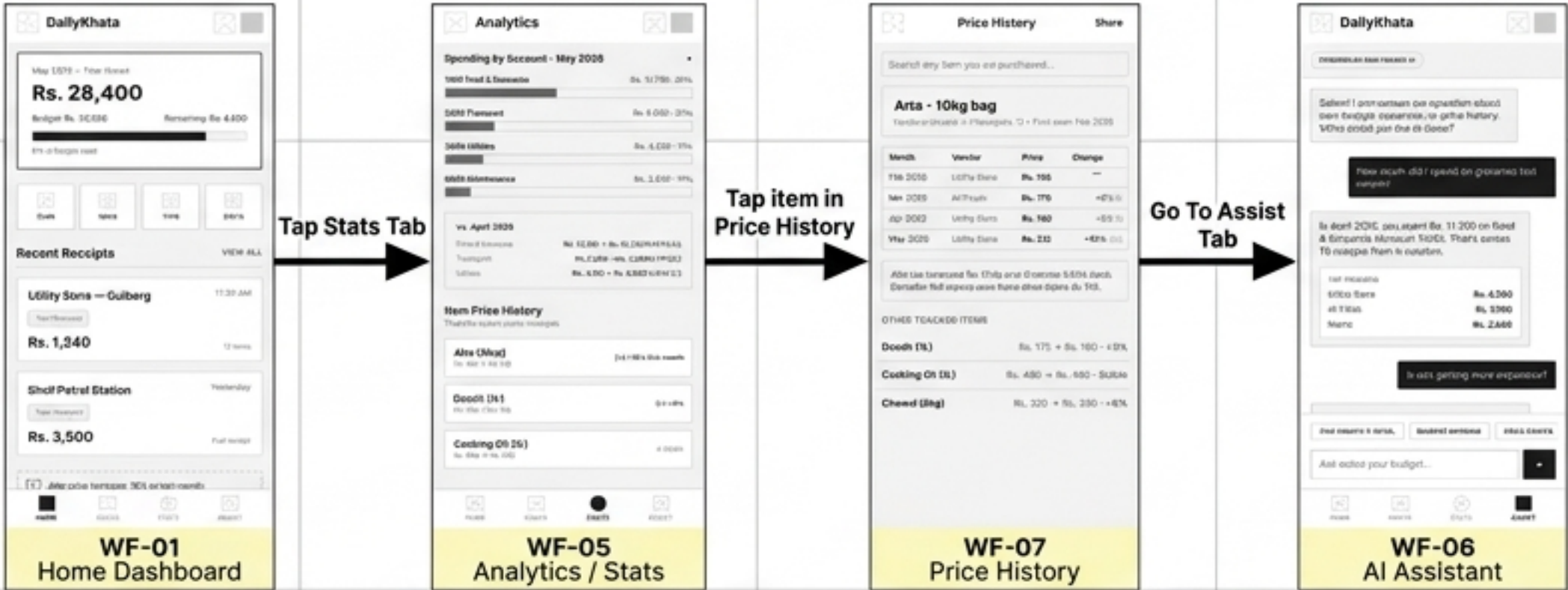
Home -> Manual Entry Form ->
Save to Ledger.



CORE NAVIGATION: HISTORICAL DATA & PROACTIVE ALERTS

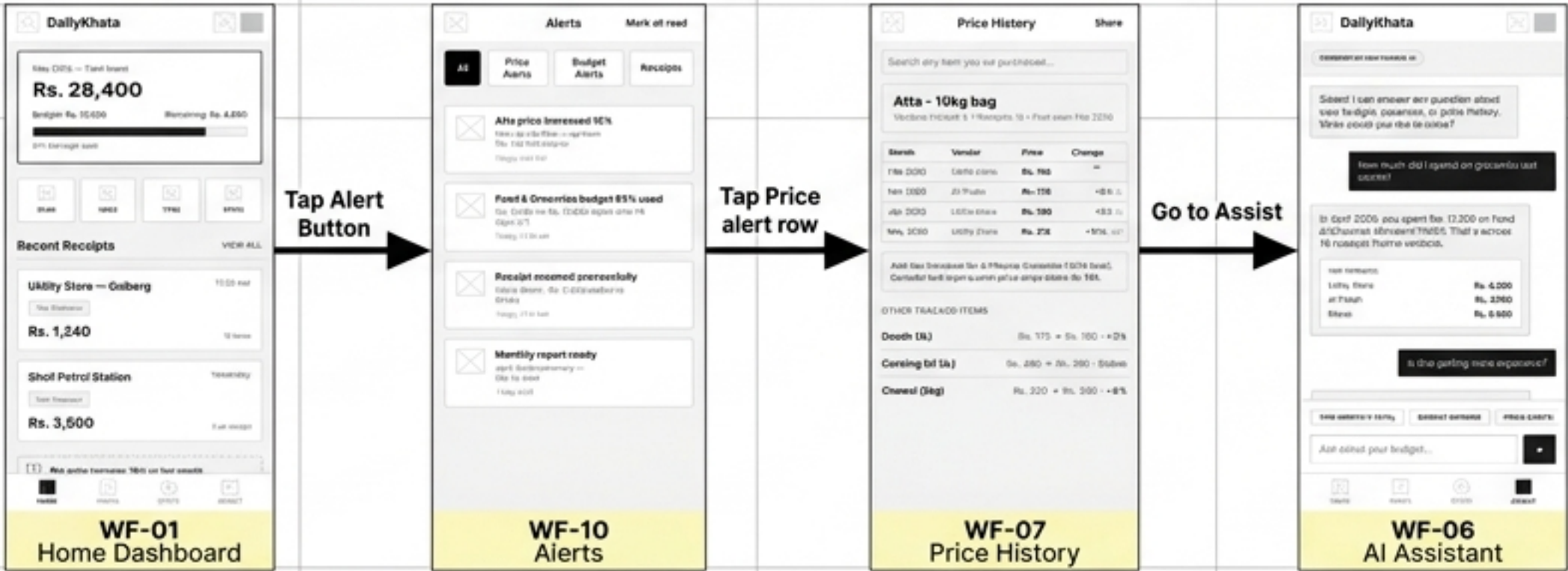
FLOW B: PRICE HISTORY (Task T2)

Analytics Tab -> Select Tracked Item
-> View Timeline -> Escalate to AI
for contextual questioning.



FLOW E: RESPOND TO ALERT (Task T6)

Notification Center -> Tap Price Alert
-> View Item History -> Ask AI for
budget adjustment recommendations.



SECTION 02

USER TEST PLAN

Empirical usability validation utilizing think-aloud protocols and the System Usability Scale (SUS).

FORMATIVE USABILITY STRATEGY

PRIMARY GOAL: Evaluate prototype navigability, clarity of AI uncertainty flags, and discoverability of conversational analytics.
TARGET SAMPLE: N=5 to 6 participants across two specific archetypes.

GROUP A: HOMEMAKERS	GROUP B: STUDENTS & FREELANCERS
Ages 28-50	Ages 18-28
Low digital literacy.	High digital literacy.
Currently relies on analogue notebooks or mental tracking.	Has tried and abandoned digital finance tools previously.
Evaluated to test the baseline accessibility and friction of the app.	Evaluated for comprehension of power-user analytics features.

THE 6 KEY USABILITY TASKS:

PERSONA-TO-TASK MATRIX

TASK	PERSONA	TARGET FLOW	OBSERVATION GOAL
T1 (Scan/Verify)	Rukhsana	Flow A	Does user understand the yellow dashed fields?
T2 (Price History)	Rukhsana	Flow B	Can they find the timeline table and read the insight card?
T3 (Manual Cash Entry)	Rukhsana	Flow C	Is the 'Maintenance' category intuitive for a plumber fee?
T4 (Ask AI)	Zaid	Flow B Variant	Do they use natural language or look for a rigid menu?
T5 (Analytics Review)	Zaid	Flow D	Is the MoM comparison card clear?
T6 (Price Alert)	Zaid	Flow E	Can they navigate from alert to actionable insight?

SESSION PROTOCOL & MODERATOR SCRIPT

00:00	00:05	00:10 TASK EXECUTION	00:40	00:45
00:00 WELCOME & CONSENT	THINK-ALOUD TRAINING	00:10 TASK EXECUTION	00:40 SUS ADMINISTRATION	00:45 DEBRIEF
Establish voluntary participation, anonymity, and recording permission.	Practice exercise using a basic calculator app to ensure participants actively narrate their internal cognitive processes.	Read standardized prompts. Strict moderator rules: No hints unless stuck for >90s (use neutral prompt: 'What would you try next?'). Max 5 mins per task.	Quantitative evaluation phase.	Open-ended discussion on expectations vs reality.

EMPIRICAL LOGGING & DATA CAPTURE

LOGGING SHEET

Task		Time	Success (Y/N/P)	Critical Incidents
T1				
T2				
T3				
T4				
T5				
T6				

QUANTITATIVE METRICS

Task Time (mm:ss) measured explicitly from prompt completion to task resolution or abandonment.

SUCCESS STATES

Classified strictly as Y (Success), N (Failure), P (Partial Success - meaning the user required a neutral prompt or experienced significant difficulty).

QUALITATIVE CAPTURE (CRITICAL INCIDENTS)

Logged sequentially. Defined as any observable event that meaningfully helps or hinders task completion.

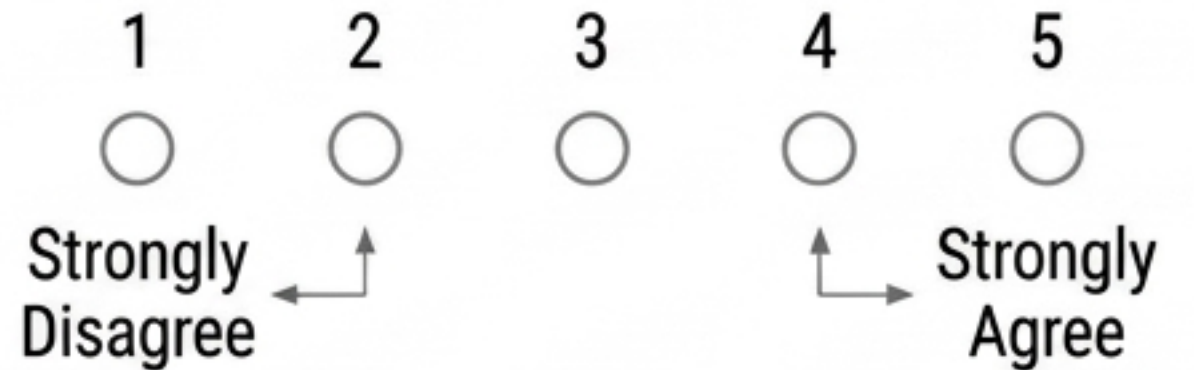
Severity Scale: 1 = Minor confusion, 3 = Significant delay, 5 = Complete breakdown/task failure.

POST-TEST EVALUATION & ETHICAL STANDARDS

SYSTEM USABILITY SCALE (SUS)

- A validated 10-item Likert questionnaire (1-5 scale) administered immediately post-tasks.

- Provides a single benchmark score (0-100) to quantify baseline satisfaction and complexity.



INFORMED CONSENT FRAMEWORK

TRANSPARENCY:

Purpose of study clearly stated to all participants.

PRIVACY:

All data strictly anonymized (e.g., users referred to as P01, P02).

AGENCY:

Participation is strictly voluntary; users retain the right to skip tasks or withdraw at any time without penalty.

MILESTONE SUMMARY & PATH FORWARD

M03 ACCOMPLISHMENTS

- Successfully translated complex user needs (identified in M02) into a cohesive, interactive 13-screen architecture.
- Established a robust, repeatable empirical testing framework ready for deployment.

IMMEDIATE NEXT STEPS

01: Execute usability sessions with N=6 participants.

02: Synthesize SUS scores and Critical Incident logs to identify UI friction.

03: Iterate on wireframe logic based on empirical user friction points.

04: Develop High-Fidelity interactive UI for the final capstone defense.

TEAM PATCHES — DAILYKHATA